TEJAS NALADALA

MACHINE LEARNING ENGINEER / EMPIRICAL RESEARCHER / HARDWARE FOUNDER

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I am 19 and study Engineering at the University of Washington. I founded R0 Systems, build machine-learning evaluation systems, and run empirical studies in embedding evaluation, agent search, reinforcement learning, medical imaging, and ocean computer vision. I have published three peer-reviewed plasma-engineering papers.

EDUCATION

University of Washington, Seattle
AUG 2025 - JUN 2028 EXPECTED

B.S. Engineering

+ Lavin Entrepreneurship Program, a selective multi-year Buerk Center cohort.
+ Coursework and independent work spanning ML, signal processing, embedded systems, optimization, and research methods.

WORK AND VENTURES

OpenTrade (YC S26)
JUN 2026 - SEP 2026

Machine Learning Engineer / first ML hire

+ Built the path from live market data to grounded LLM-generated trade cards and the personalized ranking that determined what each user saw.
+ Built factual-accuracy and regression checks for grounding, numerical consistency, temporal validity, unsupported claims, and structured output.
+ Turned impressions and downstream behavior into evaluation and training signals for a product that reached roughly 150,000 monthly users in two months.

R0 Systems
JUN 2024 - PRESENT

Founder and CEO / plasma-activated-water sanitation for post-harvest produce

+ Bootstrapped the company to six figures in revenue.
+ Architected the continuous-flow machine stack: patent-pending plasma reactor, 20 kV electronics, STM32 firmware, venturi gas-liquid transfer, process-water handling, mechanical assemblies, validation, and field deployment.
+ Published three peer-reviewed papers, filed two patents, and completed independent validation with national food-safety laboratories.
+ Demonstrated up to 3x shelf-life extension at under $0.50 per ton in energy cost across tomato and leafy-green studies.

Ingenium Naturae
MAY 2023 - JUN 2024

Research Engineer

+ Delivered custom plasma and automation machines for agri-food clients across power electronics, embedded control, fluidics, reactor design, and industrial mechanics.
+ Designed 20 kV discharge drivers, MOSFET switching stages, PID control loops, STM32 firmware, electrode and reactor geometries, Venturi injectors, and production CAD assemblies.
+ Handled specifications, architecture, component selection, fabrication, wiring, firmware, validation, acceptance testing, documentation, and handoff.

SEAL Lab, University of Washington
MAR 2025 - NOV 2025

Research Associate / PI: Prof. Alex Mamishev

+ Built a PPG drowsiness wearable with 95% motion-artifact rejection and a Navy hull-integrity piezoelectric sensor with real-time 512-point FFT at 10 kHz and acoustic-emission classification.
+ Co-authored technical sections of two funded grant proposals and surveyed 40+ papers on PPG processing and embedded anomaly detection.

CSIR-NIIST, Centre for Sustainable Energy Technologies
JUN 2024 - MAR 2025

Research Associate / PI: Dr. Suraj Soman

+ Fabricated dye-sensitized and perovskite solar cells using FTO substrates, TiO_2 photoanodes, thermal processing, ruthenium sensitizers, electrolytes, counter electrodes, and sealed assemblies.
+ Ran J-V curves, IPCE, photoinduced absorption spectroscopy, and XRD with Scherrer crystallite-size analysis.

RESEARCH

MTEB-Gym
JUN 2026 - PRESENT

Research collaborator / MTEB team

+ Co-developed MTEB-Gym, a label-free framework for ranking embedding models on datasets without human relevance annotations.
+ Worked on bidirectional judging, validation, uncertainty estimates, and failure handling for label-free rankings.
+ Contributed nine merged upstream changes across validation, uncertainty estimation, caching, deterministic parallelism, and failure handling.

RSNA Knee Abnormality Detection
JUL 2026 - PRESENT

Independent research / in progress

+ Built hierarchical 2.5D multiple-instance learning around ordered MRI slices and study-level aggregation.
+ Added patient/study splits, fold-integrity checks, leakage audits, label-quality analysis, and metric-consistency tests to the experiment path.
+ In progress.

maze-rl-baselines
FEB 2026 - APR 2026

Sole author / reproducible benchmark

+ Benchmarked PPO, DQN, and A2C against simple heuristics and behavior cloning on procedurally generated mazes.
+ The public repository includes experiment code, run records, manifests, and a result verifier.
+ Later runs outgrew the pinned manifest; artifact reconciliation is in progress before final reporting.

Connectome Architecture Benchmark
DEC 2025 - PRESENT

Independent audit and corrected protocol

+ Audited the original 757-row benchmark and found failures in trainable-component control, null-graph density, measured-data provenance, artifact binding, and CartPole evaluation.
+ Preserved and invalidated the original rows so the failure remains inspectable.
+ Implemented and tested CAB v2 with readout-only optimization, digest-bound measured connectomes, and edge/weight-matched nulls. The full corrected run is pending.

agentbreed
MAR 2026 - PRESENT

Sole author / preregistered synthetic pilot

+ Typed genome representation and nine search baselines under matched compute across 700+ runs, three domains, and 20 seeds.
+ Multi-component search beat prompt-only search (pooled dz = 1.32, p approximately 1e-14 across 60 seed-domain pairs). Search operators were statistically indistinguishable after correction.
+ The repository contains the tested optimization library, reproduction path, and preregistration for the pending real-LLM replication.

VISIONS '26 / OCEAN 411
AUG 2026

NSF OOI expedition RR2607 / R/V Roger Revelle

+ Fifteen-day operations cruise with 18 ROV Jason dives at Axial Seamount and Southern Hydrate Ridge, instrument work to 2,900 m, CTD casts, chlorophyll filtering, and oxygen, RAS, and PPS microbial-DNA sampling.
+ Built a provenance-aware multimodal expedition memory over video, sonar, hydrophone, CTD, and navigation streams.
+ Building the computer-vision, acoustic-alignment, calibration, and held-out validation pipeline.

PEER-REVIEWED PUBLICATIONS

Design of systems for plasma activated water for agri-food applications
J. Phys. D: Appl. Phys. 57(49), 493003 / 2024
Misra NN, Naladala T, Alzahrani KJ. DOI: 10.1088/1361-6463/ad77de.

Design of a continuous PAW disinfection system for fresh produce industry
Innov. Food Sci. Emerg. Technol. 97, 103845 / 2024
Misra NN, Naladala T, Alzahrani KJ, Sreelakshmi VP, Negi PS. DOI: 10.1016/j.ifset.2024.103845.

Design and construction of continuous industrial-scale cold plasma equipment
Innov. Food Sci. Emerg. Technol. 97, 103840 / 2024
Misra NN, Sreelakshmi VP, Naladala T, Alzahrani KJ, Negi PS. DOI: 10.1016/j.ifset.2024.103840.

Three peer-reviewed papers / 57 citations / h-index 3 as of Sep 2026.

MTEB-Gym
Public upstream contributions
Label-free embedding evaluation with synthetic queries, bidirectional pairwise judging, Bradley-Terry ranking, and reliability diagnostics.

Procedural-Maze RL Baselines
Artifact reconciliation in progress
Experiment code, run records, manifests, and a result verifier are public; later runs currently exceed the pinned manifest.

AgentBreed
Preregistered synthetic pilot
Seven hundred evaluations of configuration-space richness and search operators under matched compute.

Connectome Architecture Benchmark
Invalidated pilot / corrected protocol pending
The repository preserves the failed artifact and enforces the corrected CAB v2 protocol.

TECHNICAL REPORTS AND PATENT

VLM Inference Optimization
Technical report / advised by Prof. Zhuang Liu, Princeton University / Jun-Jul 2026
AlphaEvolve-style pipeline search on CharXiv chart QA with fixed Qwen3-VL-2B weights. Canonical emission and robust extraction gain 24-31 points; manual routing, gates, constrained decoding, and multi-resolution voting reach 73.0% held-out, 2.5x naive. Wilson intervals and paired McNemar p = 2.4e-6.

Single-GPU Model Serving
Technical report / advised by Prof. Juncheng Yang, Harvard University / Jun 2026
80-point vLLM matrix on one H100: dense/MoE crossover near concurrency 3, Qwen admitting 42 requests versus GLM's 10 at 49K-token generation, and a 20-point GPQA swing from token budget. An in-situ probe caught PCIe-class bandwidth before publication.

Provisional patent
Venturi-plasma integration / two additional R0 patents pending
Continuous plasma-activated-water system integration and dosing architecture.

mimic
Robotics / Mar 2026
Browser-to-robot imitation learning: WebRTC teleoperation into a MuJoCo Franka arm, demo recording, ACT or Diffusion Policy training, ONNX export, a configured 60 Hz control loop, and 108 tests.

engram
Neuromorphic learning / Apr 2026
Rust-core continual-learning runtime using local Hebbian and three-factor STDP updates, persistent memory, an action-veto layer, and PyO3 bindings.

Forge
Agent runtime / Feb 2026
Provider-agnostic multi-agent runtime across eight provider families with tools, memory, cost ceilings, FastAPI/WebSocket observability, Python and TypeScript packages, and 70 tests.

WireML
Foundation-model workbench / Apr 2026
Terminal node graph for CLIP or DINOv2 few-shot training and ONNX export across CUDA, MLX, MPS, ROCm, DirectML, and CPU.

SELECTED OPEN SOURCE

delphi-quant
Verification / Jan-May 2026
Retrospective systematic-strategy verification with transaction costs, walk-forward splits, look-ahead checks, multiple-comparison correction, synthetic reproducibility, and structured rejection logs.

RECOGNITION AND PROGRAMS

Carnegie Mellon Venture Challenge / 2nd place, national round, 150+ teams / $4,500 / Mar 2026

UW Science & Technology Showcase / Grand Prize and Best Pitch / $2,750 / Jan 2026

Red Bull Basement / 2nd place / 2026

Buerk Center Prototype Grant / $2,000 / 2026

Founders Inc pitch day presenter / Jul 2026

a16z Speedrun S008 / R0 Systems / 2026

Y Combinator F26 and Startup School 2026 / on-stage media interview, 1 of 15 among about 6,000 attendees

Z Fellows / 2026

Lavin Entrepreneurship Program / UW Buerk Center

VISIONS '26 expedition / Aug 2026

TECHNICAL STACK

Statistics and methods. Pre-registration; Bradley-Terry ranking; mixed effects; TOST equivalence; Sobol decomposition; bootstrap CIs; Welch's t; Holm and Benjamini-Hochberg correction; McNemar and Wilson intervals; MAP-Elites; multiple-instance learning; fuzz and property-based testing.

ML and serving. PyTorch; Hugging Face Transformers and PEFT; DDP/FSDP; bf16; INT4/INT8, GPTQ, AWQ; vLLM; Stable Baselines 3; Gymnasium and MiniGrid; DICOM/pydicom; RAG and ChromaDB.

Systems and hardware. Multi-node H100/H200 Slurm; Docker; AWS; FastAPI; WebRTC; STM32 and embedded C; high-voltage supplies; PID and real-time control; DSP; KiCad; Raspberry Pi; Pixhawk; CAD and fabrication.

Languages. Python (expert); TypeScript; Rust; C/C++; R; SQL; MATLAB. English, Telugu, and Hindi; conversational Mandarin.